DIGITAL FORENSIC INVESTIGATION REPORT

DFIR-001 | Apex Financial Solutions | Simulated evidence examination

Examiner: Nayanthi Weerasuriya **Examination:** 13 October 2026 **Application:** Autopsy 4.23.1

Report status: Portfolio-ready training report. Findings are limited to the documented evidence. Screenshot inventory requires local file confirmation.

1. Executive summary

A fictional corporate file-handling scenario was constructed on a 1 GB Windows virtual hard disk (VHD). Six relevant evidence items were examined: one recoverable Shift+Delete document (E01), three allocated business documents (E02-E04), and two normally deleted documents retained in the Windows Recycle Bin (E05-E06). The examination demonstrates preservation, hash verification, filesystem artifact analysis, Recycle Bin metadata decoding, and timeline reconstruction. No evidence establishes a real theft, perpetrator, or unauthorized external transfer.

2. Scope, evidence and integrity

The original evidence image was **Apex-Lab.vhd**; a preserved copy was named **Master copy.vhd** and an analysis copy **Working copy.vhd**. Initial SHA-256 values for all three were identical. The examiner subsequently rechecked the master and reported the same value; that recheck was not independently witnessed and no precise verification time is provided.

Recorded whole-image SHA-256: 34BF5F094CE7CC1977C69CF6C929F48978C4F27C19F7DFFA4EE769FD645AEFDC

These are whole-VHD hashes, not individual file hashes. The uploaded hash export documents three matching results. Treat the preserved master as evidence; analysis should use the working copy without writable mounting.

3. Methodology

Files were created and deleted in a controlled simulation; the VHD was detached and copied for preservation. Using Autopsy 4.23.1, the examiner inspected the data-source tree, allocation status, textual content, filesystem metadata, deleted-files view, and Timeline feature. Recycle Bin $R files were examined for retained content, while $I version-2 binary records were decoded for original path, size, and Windows FILETIME deletion timestamps. Observed times are expressed in UTC+05:30 (Sri Lanka local time).

4. Findings register

ID	Evidence / state	Key observations
E01	Internal_Budget_Fictional.txt — unallocated	686 bytes; deleted-file text readable; created 21:17:41; NTFS Changed 21:21:36 (not proven deletion time).
E02	Client_List_Fictional.csv — allocated	258 bytes; readable fictional client data; all four filesystem times 21:19:57.
E03	Q3_Revenue_Fictional.csv — allocated	173 bytes; readable fictional revenue data; all four filesystem times 21:20:39.
E04	Meeting_Notes.txt — allocated	699 bytes; created 21:18:04; modified 21:21:27; added follow-up sentence present.

ID	Evidence / state	Key observations
E05	$RAMTAZN.txt / $IAMTAZN.txt — Recycle Bin	Readable client content; original F:\Confidential\Client_List_Fictional.txt; $I deletion 21:20:06.793.
E06	$RF6G3QU.txt / $IF6G3QU.txt — Recycle Bin	Readable revenue content; original F:\Confidential\Q3_Revenue_Fictional.txt; $I deletion 21:20:43.469.

5. Detailed assessment of deletion artifacts

E01 - Shift+Delete: Autopsy listed the internal budget document as deleted with unallocated filename and metadata entries; the original fictional budget text remained readable. The recorded NTFS metadata-change time (21:21:36) is consistent with the simulation log stating deletion at approximately 21:22, but it is not a direct timestamp of deletion.

E05 - ordinary Delete: The allocated $R content file retained 258 bytes of readable client-list material. The paired $I file identified **F:\Confidential\Client_List_Fictional.txt**, size 258 bytes, and a deletion time of **21:20:06.793**. The $R filesystem metadata reports creation at 21:16:28 and Changed at 21:20:06; those are distinct from $I FILETIME.

E06 - ordinary Delete: The allocated $R content file retained 173 bytes of readable revenue material. The paired $I file identified **F:\Confidential\Q3_Revenue_Fictional.txt**, size 173 bytes, and a deletion time of **21:20:43.469**. The $R creation time was 21:16:57 and Changed was 21:20:43.

The $I timestamps were decoded from eight-byte little-endian FILETIME values (100 ns intervals since 1601-01-01 UTC), then converted to UTC+05:30. Original paths were interpreted from UTF-16LE strings. These are Recycle Bin events, not proof that data left the device.

6. Selected chronological timeline

Local timestamp	ID	Observed event / source
21:16:28	E05	Created: Client_List_Fictional.txt ($RAMTAZN.txt) ($R file system metadata)
21:16:57	E06	Created: Q3_Revenue_Fictional.txt ($RF6G3QU.txt) ($R file system metadata)
21:17:41	E01	Created: Internal_Budget_Fictional.txt (NTFS Created)
21:18:04	E04	Created: Meeting_Notes.txt (NTFS Created)
21:19:57	E02	Created: Client_List_Fictional.csv (NTFS Created)
21:20:06.793	E05	Moved to Recycle Bin: Client_List_Fictional.txt ($IAMTAZN.txt FILETIME)
21:20:39	E03	Created: Q3_Revenue_Fictional.csv (NTFS Created)
21:20:43.469	E06	Moved to Recycle Bin: Q3_Revenue_Fictional.txt ($IF6G3QU.txt FILETIME)
21:21:27	E04	Modified: Meeting_Notes.txt (NTFS Modified)
21:21:27	E04	Metadata changed: Meeting_Notes.txt (NTFS Changed)
21:21:36	E01	Metadata changed: Internal_Budget_Fictional.txt (NTFS Changed)

Full event-level timestamps, including coincident modification and change events, are provided in **Evidence_Timeline_Corrected.csv**.

7. Simulation log comparison

The contemporaneous activity log records a meeting-notes edit at approximately 21:21 and a budget-file deletion at approximately 21:22, consistent with the observed meeting-notes modified time (21:21:27) and the budget metadata-change time (21:21:36), respectively. The latter remains circumstantial and does not establish precise deletion time.

The log describes deleting earlier .txt versions and recreating them as .csv files. Detailed $I metadata establishes when the .txt originals were recycled. Notably, the replacement CSV creation times (21:19:57 and 21:20:39) precede corresponding $I deletion times (21:20:06.793 and 21:20:43.469). Thus the record supports a replacement workflow, but not a strict delete-before-recreate sequence.

8. Limitations and quality controls

This was a staged educational exercise. The report does not identify a real data breach, transfer destination or actor. Filesystem time semantics vary, and the absence of an artifact is not evidence of secure erasure. Visible text recovery is not independent byte-for-byte validation of exported individual files. No host authentication logs, USB-connection history, or external-transfer telemetry were collected. Screenshot filenames below are based on the case notes; confirm each file exists before making the repository public.

Screenshot references: E01_Deleted_File_Listing.png; E01_Deleted_File_Content.png; E01_Deleted_File_Metadata.png; E02_Client_List_Content.png; E02_Client_List_Metadata.png; E03_Revenue_Content.png; E03_Revenue_Metadata.png; E04_Meeting_Notes_Content.png; E04_Meeting_Notes_Metadata.png; E05_Recycle_Bin_Metadata.png; E06_Recycle_Bin_Metadata.png; TL01_Forensic_Timeline.png. Preserve a screenshot of each $I hex view if available.

9. Conclusion and recommended next actions

The evidence demonstrates recovery of the simulated deleted budget file, confirmation of allocated documents and a later meeting-notes modification, and recovery of two ordinarily deleted document versions through paired NTFS Recycle Bin artifacts. The case provides a reproducible introductory example of disk forensics, metadata validation and cautious interpretation. Before GitHub publication, verify screenshot filenames, retain the hash manifest and simulation log, redact any incidental identifiers, and keep large VHD images out of the public repository.